

Dr. Babasaheb Ambedkar Technological University, Lonere -Raigad

Gourishankar Educational Charitable Trust

Gourishankar Institute of Pharmaceutical Education and Research Limb, Satara.

End Semester Theory Examination WINTER-2024

Semester: I

Class: First Year B. Pharm.

Subject: Remedial Math's (BP106RMT)

Time: 2:00 hour

Date:- 04/01/2025

Total marks:- 35

Q. No. 1 Attempt any one

(10 marks)

A) Solve integral $x^7 e^x$

B) Verify clayey Hamilton theorem

$$\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 2 & 2 \end{pmatrix}$$

Q. 2 solve any 5

(25 marks)

1) Solve the equation $X+Y+Z=3$, $2x-y+z=-1$, $2x+y-3z=-9$ by crammer rule

2) $F(x) = (x-3)/4x-5$, $g(x) = (3+5x)/(4x-1)$ then show that $f \circ g(x) = x$

3) Find Laplace transforms of $6e^{-3t} - 2\cos t + 5t^3 + 2$

4) Partial fraction $x/(x-1)(x-2)$ ($x=3$)

5) Show that the point A(1,7) B(4,2) C(-1,-1) and D(-4,-4) are vertices of square

6) Solve partial fraction I) $2x/(x-1)(x-2)(x-3)$ II) $2x/(2x-1)(3x-1)(4x-1)$